



A spectral difference preservation network based on Mamba pyramid for hyperspectral image compression

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ARTICLE INFO

Keywords:

Hyperspectral image compression

Frequency domain features

Mamba

Deep learning

Rate distortion

ABSTRACT

In lossy hyperspectral image compression, reconstructed images often suffer from block or blur effects. This phenomenon is even more pronounced at high compression ratios. Low quality frequency domain features and coarse feature extraction are important factors for this phenomenon. In this paper, a spectral difference preservation network based on Mamba pyramid for hyperspectral image compression (SDMNet) was proposed. Firstly, a spectral difference preservation head (SDPH) was designed, integrating grouped element-wise reconstruction and complex frequency domain feature analysis, to retain high-quality frequency domain features. Secondly, Mamba was introduced into the field of hyperspectral compression for the first time. A Mamba pyramid feature enhancement module (MPFM) was developed, incorporating dimensional squeeze and pyramid Mamba for refined feature extraction. Finally, based on the above innovations, this paper proposes an efficient SDMNet in conjunction with the Variational Autoencoder (VAE) framework for hyperspectral image compression. Some experimental results show that on the dataset Pavia Centre, Chikusei and Houston, compared with some advanced compression models, SDMNet has consistently demonstrated excellent compression performance. Especially on dataset Pavia Centre, the PSNR obtained by SDMNet is increased by up to 11 %. The related code will be released at <https://github.com/shikaijieskj/SDMNet>.

1. Introduction

Hyperspectral images can reveal the characteristics of a variety of ground objects, including landforms, temperatures, and crop species. Therefore, hyperspectral imagery boasts diverse applications spanning multiple disciplines, among them environmental assessment, geological investigation, and military surveillance [1–3]. In addition, with advances in sensor technology [4,5], the resolution continues to increase, and countless hyperspectral images, numbering in the billions, are taken and sent across various platforms [6,7]. Therefore, it is imperative to promptly advance hyperspectral image compression techniques that offer enhanced compression efficiency while maintaining high image fidelity.

At present, some achievements have been made by traditional image compression techniques [8,9]. The classical JPEG [10] and JPEG2000 [11] technical architectures mainly include image transformation, quantization processing, and entropy encoding. It first transforms image

and removes redundant information. Subsequently, important information is filtered and retained through quantitative operations. Finally, the data with weakened correlation is further compressed with the help of entropy coding techniques. In addition, better portable graphics (BPG) [12,13] and WebP [14] have also come into existence as the times require. However, the traditional compression method has a limitation, that is, when pursuing a high compression ratio, the reconstructed image will have a significant blurring effect. Therefore, for hyperspectral images with rich features, it is difficult for traditional compression methods to achieve high-fidelity image reconstruction.

In search of technological breakthroughs, researchers have turned attention to deep learning technologies, which have become popular in the last few years [15–17]. Classical deep learning-based image compression models include Autoencoder (AE) [18,19] and Variational Autoencoder (VAE) [20,21]. Liu et al. constructed an underwater image compression scheme, which utilizes AE for extremely low bit rate compression. Dual-path coding is used to achieve high-quality

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<https://doi.org/10.1016/j.patcog.2026.113088>

Received 8 July 2025; Received in revised form 12 November 2025; Accepted 11 January 2026

Available online 14 January 2026

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compression [19]. Xu et al. constructed a VAE-based convolutional neural network (CNN) for compression. It uses a linear CNN and a nonlinear activation function to improve performance [21]. It is worth mentioning that, compared to AE, VAE has a stronger data modeling capability. VAE learns a continuous latent space, replacing the discrete latent representation generated by AE. This continuity allows every point in the latent space to have significance, thereby supporting smoother interpolation and the generation of new data. Over the past few years, baseline networks grounded in VAE have demonstrated outstanding capabilities in image compression, outperforming traditional approaches and attaining superior compression quality [22,23]. These networks grounded in VAE typically contain encoders, entropy encoding modules, and decoders. They first use neural networks to compress the image, then quantize the data, and finally use traditional encoding techniques to generate efficient bit streams. Furthermore, to enhance the precision of modeling and leverage prior knowledge effectively, certain models incorporate diverse entropy models into their frameworks. These include the Laplace entropy model, hybrid Gaussian model, and factorized entropy model [24,25]. Building upon these theoretical underpinnings, researchers have devised a range of deep learning-based compression networks tailored for remote sensing imagery, yielding favorable rate-distortion outcomes [26,27]. Although these methods can achieve image compression, the reconstructed image at high compression ratio will still be blurry. The proposed SDMNet achieves clear reconstruction under high compression ratio by retaining high-quality frequency domain features and refining feature extraction.

Three primary categories of deep learning models are employed for the hyperspectral image compression: CNN-based approaches [26,27], Transformer-based approaches [28,29], and contrastive learning-based approaches [30]. In the CNN-based approaches, Zhou et al. constructed a hyperspectral image data compression method for lightweight airborne systems. This method uses an optimized decoder combined with a channel attention to recover data and achieves high-quality reconstruction [26]. In the Transformer-based approaches, Zhang et al. proposed an efficient hyperspectral image compression network that adopts the CNN-Transformer Hybrid (CTM) architecture. To meet the requirements of the receptive field of the band data obtained by the spaceborne pushbroom hyperspectral sensor, a non-square convolutional layer and CTM blocks were designed to construct a lightweight encoder [29]. It should be noted that Transformers exhibit quadratic time complexity when processing long sequences and high-dimensional data, leading to sharply increased computational and memory demands. In contrast, Mamba is based on a selective state space model (SSM) and uses linear time complexity, enabling dynamic selection of relevant information for computation, which improves efficiency and reduces unnecessary calculations. Therefore, our paper introduces Mamba technology to the field of hyperspectral compression for the first time. In the contrastive learning-based approaches, Guo et al. constructed a high-quality hyperspectral image compression network. It extracts and organizes discriminant attributes from the original hyperspectral image through comparative information coding and enhances the discriminant nature of latent features under different channel indexes [30]. Although the aforementioned approaches attain good compression outcomes, they still lose a lot of detailed features at high compression ratios. This is because of the loss of features in the frequency domain and the rough feature extraction method. This also results in suboptimal performance of these methods.

Hyperspectral images contain abundant information in the frequency domain. High-quality frequency domain features facilitate the extraction of edge and texture structures. However, the current learning-based hyperspectral image compression network usually directly fuses the information of the bands, ignoring the differences between the bands. This also leads to the loss of many features in the frequency domain, which makes the reconstruction image blurry at high compression ratios. Therefore, efficient alignment of the frequency domain features has become a serious challenge to be solved in

hyperspectral image compression.

To mitigate the aforementioned problem, a spectral difference preservation network based on Mamba pyramid for hyperspectral image compression (SDMNet) is proposed. The network not only retains high-quality frequency domain features but also realizes refined feature extraction. Hyperspectral images contain abundant frequency-domain features, which facilitates the alignment of edge and texture features. Therefore, a spectral difference preservation head (SDPH) is constructed to align the frequency-domain features. Secondly, hyperspectral images contain multiple categories of features, including spectral, irregular, and multi-scale features. Therefore, a Mamba pyramid feature enhancement module (MPFM) was constructed to extract multiple features in a refined manner. Finally, this paper proposes an efficient hyperspectral compression model (SDMNet) in conjunction with the VAE.

The contributions can be summarized as follows:

- (1) Mamba technology was introduced into the field of hyperspectral image compression for the first time, which improved the model's ability to capture long-range features, and effectively improved compression efficiency and image fidelity. This introduces a new technical perspective to the field of hyperspectral data compression.
- (2) A spectral difference preservation head (SDPH) is designed, which consists of grouped element-wise reconstruction header (GERH), complex frequency domain feature analysis block (CFB) and multi-scale feature interaction block (MIB) to efficiently align the frequency-domain features in the original hyperspectral image and the reconstructed hyperspectral image.
- (3) A Mamba pyramid feature enhancement module (MPFM) is designed through a dimensional squeeze block (DSB), a pyramid Mamba block (PMB), and a pointwise convolution feature enhancement block (PFB) to achieve refined extraction of multiple features.
- (4) In this study, SDPH, MPFM, and VAE frameworks were efficiently embedded, and a spectral difference preservation network based on Mamba pyramid for hyperspectral image compression (SDMNet) was constructed. Many experiments on three public hyperspectral image datasets have proved the superior compression performance of SDMNet.

The remaining sections of this paper are organized as follows. In Section II, A detailed explanation of SDMNet and its individual modules is provided. In Section III, relevant datasets are presented. Through several evaluation indicators, SDMNet and advanced compression methods are analyzed and compared in detail. Section IV delves into the conclusions drawn and explores potential avenues for future research.

2. Methodology

In the section, SDMNet, SDPH, MPFM, and rate distortion optimization strategies, are described in detail

2.1. The Overall framework of the proposed SDMNet

From the perspective of aligning the frequency domain features in the original hyperspectral data and the reconstructed hyperspectral data, the proposed SDMNet retains the differences between the spectrums and realizes the refined extraction of various features. Efficient hyperspectral image compression is achieved through SDPH which retains high-quality frequency domain features, MPFM which retains refined feature extraction, and VAE framework that can preserve detailed features. SDPH mainly involves three sub-modules, namely GERH, which can preserve the difference between bands, CFB, which can retain edge features, and MIB, which can enhance the interaction of multi-scale features. It should be noted that when the function of SDPH is compression, it contains a downsampling block. When the function of SDPH is reconstruction, it contains an upsampling module. This will be explained in detail later. MPFM mainly involves three sub-modules,

namely DSB, which can compact spectral features, PMB, which can selectively extract important features, and PFB, which can enhance feature-by-element features. VAE mainly involves some downsampling modules, upsampling modules, and probabilistic models for modeling. Finally, all components are efficiently trained under the guidance of the rate distortion optimization strategy, and a high-quality compression model is obtained.

Fig. 1 shows SDMMNet. C block1-3 compresses the data mainly by decreasing the spatial dimensions while expanding the spectral dimensions. R block1-3 reconstructs the data mainly by enlarging the spatial dimensions and decreasing the spectral dimensions. Probability models are primarily used for probabilistic modeling and further compression. The data between AE and AD is a compressed bit stream. For more details of C blocks 1-3, R blocks 1-3, Hyper encoder, and Hyper decoder, please refer to Tables 1 and 2 in the appendix.

The data processing process for SDMMNet is as follows. Compression: First, the frequency domain features of the hyperspectral data block are extracted through SDPH, and down-sampled to obtain a preliminarily compressed data block. Then the data block is further compressed through C block1 and C block2 and input into MPFM for refined feature extraction. The refined features are then passed through C block3 to obtain a deep feature. And the feature is further removed of statistical redundancy through Q, AE and probability model to obtain a compressed bitstream. Reconstruction: The model combines the bitstream with the probability model, and reconstructs the hyperspectral data with high quality through AD, R block3, MPFM, R block2, R block1 and SDPH.

2.2. Spectral difference preservation head

Hyperspectral image data differs from RGB image data in that it typically has hundreds or thousands of bands. However, there are significant differences between hyperspectral image bands, resulting in

abundant frequency domain features. Fig. 2(a) and Fig. 2(b) show the Pearson correlation coefficient matrix and spectral curves on the hyperspectral dataset Chikusei, respectively. It is observable that the similarity between the different bands is very low, which strongly proves the large differences within the spectrum. This difference is noted as dissimilar features. However, the current hyperspectral image compression methods usually directly reduce and fuse the spectral dimension, which greatly loses such dissimilar features, resulting in the suboptimal rate distortion performance of the model. Therefore, an SDPH is constructed for extracting and reconstructing rich frequency-domain features. SDPH mainly consists of three sub-modules, namely GERH for preserving dissimilar features between bands, CFB for preserving texture features, and MIB for enhancing multi-scale feature interaction.

SDPH is a module that aligns the frequency domain features between the original data and the reconstructed data. The module is used in both compression and reconstruction networks, but there are subtle differences. As shown in Fig. 3, when the SDPH module is in a compression network, it contains an additional Downsample module to elevate the channel dimension. However, when SDPH is in the reconstruction network, this downsampling module is replaced with an upsampling module to reduce the channel dimension. Due to structural symmetry, only SDPH for compression networks is introduced here.

Hyperspectral data will sequentially pass through GERH, CFB, and MIB. Specifically, data enter the SDPH and are first passed through GERH to preserve the dissimilar features between the spectra. The input data is first sliced into five sets of spectra. It should be noted that this study adopts a sequential continuous partitioning method, grouping the input spectral bands according to their original order. Specifically, every 10 consecutive bands are divided into one group. For example, for data containing 50 bands, the channels will be systematically divided into 5 groups, with channels 1 to 10 forming the first group, channels 11 to 20 forming the second group, and so on. After that, the five sets of spectra

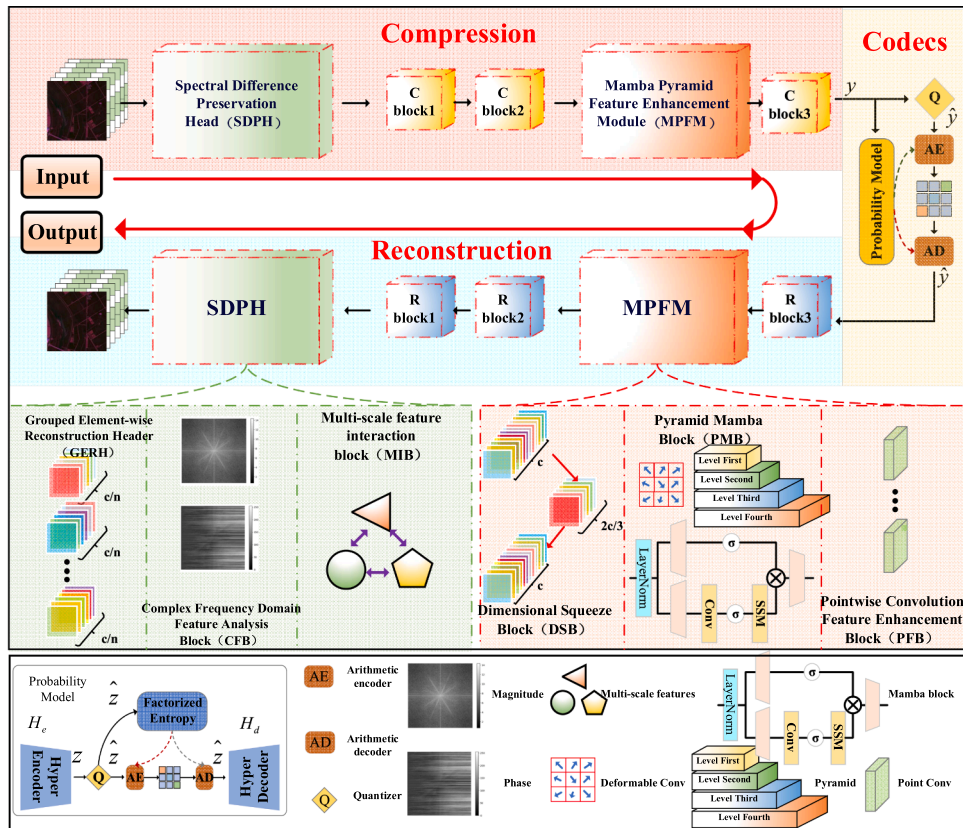


Fig. 1. The structure diagram of SDMMNet.

Table 1

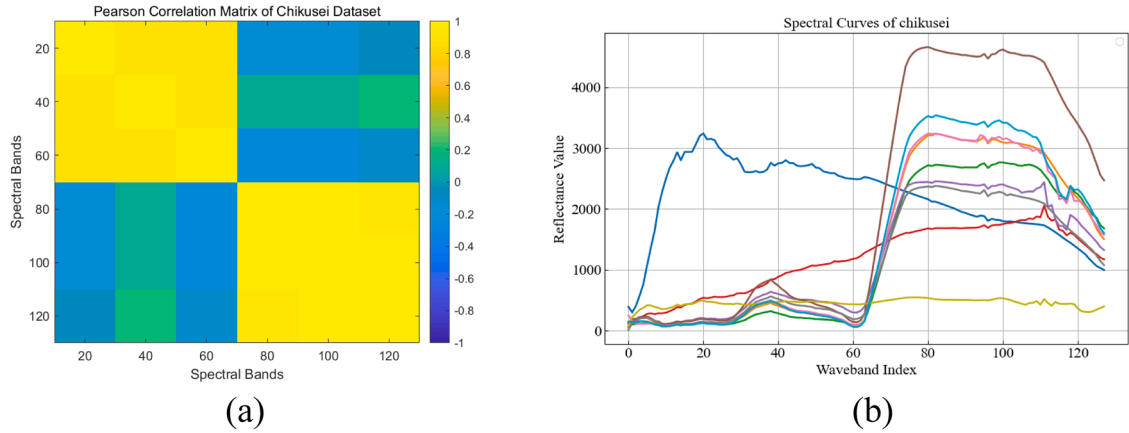
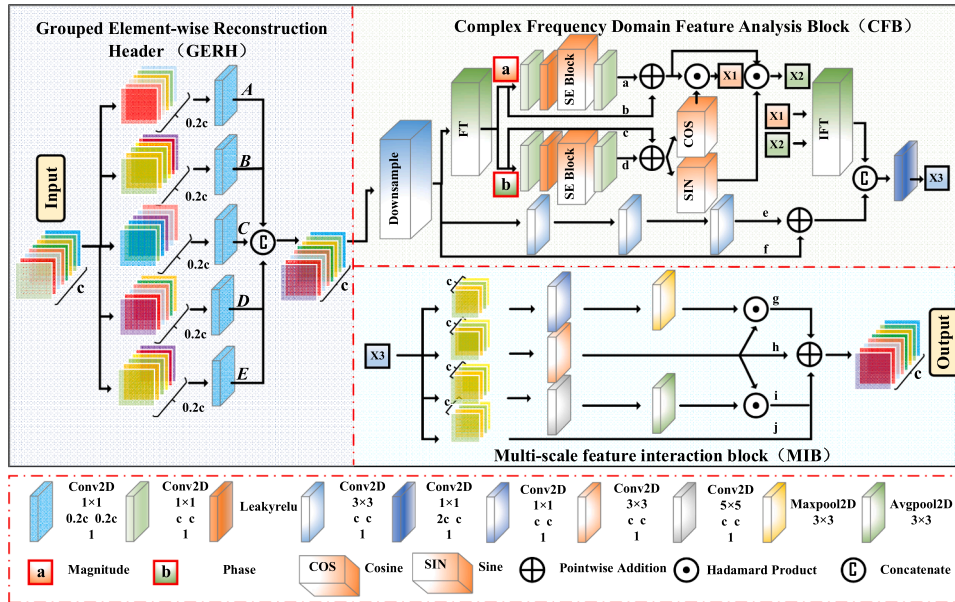
Performance comparison under high compression ratio conditions on Pavia Centre.

	Joint (bpp:0.26)	Scale (bpp:0.33)	Factorized (bpp:0.27)	MDSNet2024 (bpp:0.29)	GDRNet2024 (bpp:0.33)	SDMNet (bpp:0.25)
PSNR	33.06	36.51	32.33	35.82	35.42	36.93
MS-SSIM	5.54	8.03	3.19	7.78	7.21	9.11
SAM	9.05	7.72	8.46	7.86	8.17	7.48

Table 2

Performance comparison under high compression ratio conditions on chikusei.

	Joint (bpp:0.26)	Scale (bpp:0.30)	Factorized (bpp:0.27)	MDSNet2024 (bpp:0.29)	GDRNet2024 (bpp:0.26)	SDMNet (bpp:0.28)
PSNR	41.48	40.88	41.52	39.89	39.91	41.77
MS-SSIM	10.38	9.60	10.40	9.05	8.73	10.69
SAM	3.54	3.81	3.56	3.95	3.98	3.48

**Fig. 2.** Schematic diagram of spectral differences (a) represents the Pearson correlation matrix and (b) represents the spectral curve.**Fig. 3.** The structure of SDPH (Here, Conv2D 1×1 0.2c 0.2c 1, Conv2D represents two-dimensional convolution, 1×1 stands for the shape of the kernel, c stands for channels, and 1 stands for groups. FT stands for Fourier transform. IFT stands for inverse Fourier transform. The weights a, b, c, d, e, f, g, h, i, j are 1, 1, 1, 1, 1, 0.0625, 0.4375, 0.0625, 0.4375 respectively).

enter the five sets of point convolutions with unshared weights for element-by-element feature extraction. Here we choose five sets of point convolutions with non-shared weights to better preserve the spectral differences. It is worth mentioning that distant spectra usually exhibit

significant differences, while adjacent spectra have a certain degree of similarity. If conventional convolution (with 1 group) is used for feature extraction, all spectral features will be fused and weighted together, thereby weakening the differences between spectra. Conversely, if the

number of groups is too high (for example, 50 groups), features are extracted from each band individually, which would destroy the similarity between adjacent spectra. Therefore, this paper chooses a moderate number of 5 groups to balance both the differences and similarities between spectra. The working process of GERH can be expressed as:

$$X_{GERH} = F_{concat}(A, B, C, D, E) \quad (1)$$

Here, A, B, C, D, E stand for the five sets of tensors obtained by point convolution after the input tensors are sliced along the channel dimension. $F_{concat}(\cdot)$ stands for the functional function that stitches together multiple tensors. X_{GERH} represents the output of GERH.

Then, X_{GERH} is downsampled and input into the CFB module for texture feature extraction. CEB adopts a dual-branch structure. The upper branch is denoted as branch A. The lower branch is denoted as branch B. The main function of branch A is to transform hyperspectral data from the spatial domain to the frequency domain using Fourier transform. The quality of edge and texture features is enhanced by strengthening the interaction between amplitude and phase. Branch B is a conventional residual block. Branch A can be represented as

$$X_a, X_b = F_C(F_{FT}(X_{Downsample})) \quad (2)$$

Here, $X_{Downsample}$ represents X_{GERH} that has been subsampled, F_{FT} represents the Fourier transform function, F_C represents the function for calculating amplitude and phase, X_a represents amplitude, X_b represents

$$X_Y = \text{Sigmoid}(\text{ConvUp}(\text{ConvDown}(\text{ConvUp}(\text{ConvDown}(\text{Input})))) \odot b(\text{Input}) + a(\text{ConvUp}(\text{ConvDown}(\text{ConvUp}(\text{ConvDown}(\text{Input})))) \odot \quad (6)$$

phase.

$$X_1, X_2 = F_{Interaction}(X_a, X_b) \quad (3)$$

Here, $F_{Interaction}$ represents some feature extraction and interaction modules, such as convolutional blocks, SE blocks, COS and SIN. X_1 and X_2 represent the new amplitude and phase respectively.

$$X_{BranchA} = F_{IFT}(X_1, X_2) \quad (4)$$

Here, F_{IFT} represents the inverse Fourier transform function, $X_{BranchA}$ represents the output of branch A.

The result of CFB can be represented as

$$X_3 = \text{Conv}_{1 \times 1}(F_{Concat}(X_{BranchA}, X_{BranchB})) \quad (5)$$

Here, $\text{Conv}_{1 \times 1}$ represents point convolution, which is mainly used to fuse the features of two branches.

Finally, X_3 is input into the MIB module for multi-scale feature interaction. The MIB module adopts a four-branch structure. The first three branches add convolutions of different sizes to extract multi-scale features. In addition, two pooling layers are introduced in the first three branches to extract attention matrices for feature interaction. The last branch is similar to a residual structure and is used to accelerate network calculations. The weights of the four branches are 0.0625, 0.4375, 0.0625, and 0.4375, respectively, to reasonably fuse the features of each branch and avoid the explosion of pixel values. It is worth mentioning that the sum of the weights of these four branches is 1. This is mainly to prevent model instability caused by sudden increases or decreases in values during data processing. In addition, the weights of the second and fourth branches are significantly larger than those of the first and third branches. This is primarily because the second branch is the main

branch, and the convolution kernel size used in this branch is consistent with the backbone network. Increasing the weight of this branch is beneficial for model stability. The fourth branch, on the other hand, is similar to a residual structure and contains more original features, so its weight is also relatively larger.

2.3. Mamba pyramid feature enhancement module

Hyperspectral data often contains multiple categories of features, such as spectral features, irregular features, multi-scale features, etc. If a unified method is used for feature extraction, the diversity of features will be lost. Therefore, MPFM is proposed for the refined extraction of multiple categories of features. MPFM mainly consists of three sub-modules, namely DSB, which can compact spectral features, PMB, which can selectively extract important features, and PFB, which can enhance features element by element.

When hyperspectral data is input into the MPFM, it passes sequentially through the DSB, PMB, and PFB. Specifically, DSB will first optimize the spectral features. Hyperspectral data usually contains hundreds of bands, which leads to rich spectral features. However, these features are sparse, which hinders the process of feature extraction. Therefore, DSB compacts the spectral features by reducing and stretching the dimensions, which can effectively improve the ability to understand complex channel features. DSB can be represented as

Here, ConvDown represents the convolution that reduces the dimension, ConvUp represents the convolution that increases the dimension, $\odot$ represents Hadamard product. a and b stand for weights, which are 0.5 and 0.5 respectively, X_Y represents the output feature. Fig. 4

Secondly, X_Y is fed into PMB for selective feature extraction. Here, deformable convolution, Mamba block, and pyramid convolution are introduced to extract features in a refined manner. In hyperspectral images, many features come from the superposition of multiple objects and contain many irregular shapes. These features have different shapes, and it is challenging for traditional square convolution kernels to capture them effectively. Consequently, deformable convolution is incorporated to augment the capacity for capturing intricate features. Through learning offset values, the size and position of the convolutional kernel can be dynamically modified in response to the input, thereby enabling adaptive adjustments to accommodate the diverse shapes of various objects. Here the output of the deformable convolution is denoted as X_D . This is followed by a pyramid Mamba, which is mainly used for selective extraction of multi-scale features. It should be noted that, considering the number of parameters and efficiency, a very deep pyramid level was not chosen here; instead, four layers were used. The state space dimension of Mamba is 16, which was determined to be a better setting through parameter tuning and testing. The selective state-space model (SSM) used in the Mamba module here can dynamically determine which information to propagate or forget based on the input content. This enables an intelligent judgment of the 'importance' of multi-scale features. This 'selective' mechanism is the core step for efficiently filtering information. And parameters in Mamba are much less than that of Transformer. In this module, the comparison of parameters

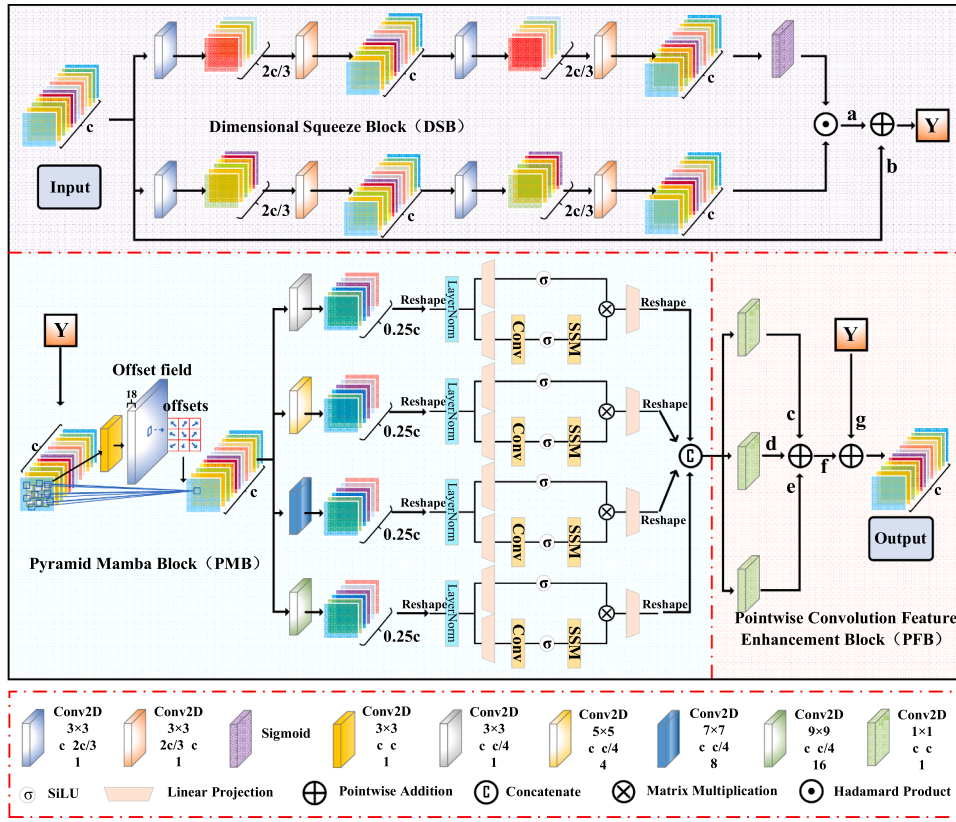


Fig. 4. The structure of MPFM. The weights a, b, c, d, e, f , and g are 0.5, 0.5, 0.33, 0.33, 0.33, 0.5, and 0.5 respectively. The input dimension, state space dimension, convolution kernel size, and intermediate layer expansion factor of Mamba are 48, 16, 4, and 2, respectively. And Mamba uses RMSNorm for normalization.

Table 3

Performance comparison under high compression ratio conditions on houston.

	Joint (bpp:0.43)	Scale (bpp:0.42)	Factorized (bpp:0.52)	MDSNet2024 (bpp:0.61)	GDRNet2024 (bpp:0.38)	SDMNet (bpp:0.44)
PSNR	41.87	41.33	41.60	42.06	41.34	42.12
MS-SSIM	12.38	11.40	11.70	12.42	11.52	12.79
SAM	1.37	1.40	1.36	1.31	1.39	1.30

and operational efficiency between Mamba and Transformer can be referenced in Table 3 of the appendix. Therefore, Mamba can extract important multi-scale features in the image with interest under low parameters. PMB can be represented as

$$X_{PFB} = f(c(\text{Conv}_{1 \times 1}(X_{PMB})) + d(\text{Conv}_{1 \times 1}(X_{PMB})) + e(\text{Conv}_{1 \times 1}(X_{PMB})) + g(X_Y) \quad (8)$$

Here, c, d, e, f, g are the weights of each branch, which are 0.33, 0.33, 0.33, 0.5, and 0.5 respectively. X_{PFB} stands for the output of PFB (i.e., the

$$X_{PMB} = F_{Concat}(\text{Mamba}(\text{Conv}_{3 \times 3}(X_D)), \text{Mamba}(\text{Conv}_{5 \times 5}(X_D)), (\text{Mamba}(\text{Conv}_{7 \times 7}(X_D)), (\text{Mamba}(\text{Conv}_{9 \times 9}(X_D)))) \quad (7)$$

Here, $\text{Conv}_{3 \times 3}, \text{Conv}_{7 \times 7}, \text{Conv}_{9 \times 9}, \text{Conv}_{5 \times 5}$ represent a set of pyramid convolutions, Mamba represents Mamba block, X_{PMB} represents the output of PMB.

Finally, X_{PMB} is input into PFB for element-by-element feature enhancement. Here, three point convolutions with unshared weights are selected. At the same time, the output of DSB is also introduced here as a residual between modules to enhance the interactivity between features. PFB can be represented as

output of MPFM).

2.4. Rate-Distortion optimization

The primary objective of training the compression algorithm is to establish an ideal balance between compression efficiency and signal fidelity. Therefore, compression frameworks generally integrate rate distortion optimization strategies. It enables efficient guidance of the training process by dynamically adjusting the encoding parameters. In short, the compression ratio is maximized while maintaining the ability

to represent the data; Reconstruction distortion is minimized under the given bitrate constraints. It can be represented as

$$\text{argminLoss} = R + \lambda D \quad (9)$$

Here, R stands for rate. D stands for distortion. Penalty coefficient λ is used to control different bitrates

$$R = R_{\hat{y}} + R_{\hat{z}} \quad (10)$$

$$R_{\hat{y}} = -\sum_i \log_2(p_{\hat{y}}(\hat{y})) \quad (11)$$

$$R_{\hat{z}} = -\sum_i \log_2(p_{\hat{z}}(\hat{z})) \quad (12)$$

Here, $p_{\hat{y}}$ and $p_{\hat{z}}$ are the density function of quantized latent representation and hyper-latent representation, respectively.

$$D = \frac{1}{m} \sum_{i=1}^m (\hat{X} - X)^2 \quad (13)$$

Here, X represents the original image, $\hat{X}$ represents the reconstructed image, and D represents the mean squared error between them, that is, the distortion.

3. Experimental results and analysis

Extensive experiments were conducted on several hyperspectral image datasets, including Pavia Centre, Chikusei and Houston, which contained a variety of ground object and could effectively evaluate these models. In addition, five advanced image compression algorithms were chosen for comparison, namely Joint [31], Scale [32], Factorized [32], MDSNet2024 [33], and GDRNet2024 [34]. Results show that GEMNet achieves the best performance on all datasets in peak signal-to-noise ratio (PSNR), multi-scale structural similarity index measurement (MS-SSIM) and spectral angle mapping (SAM).

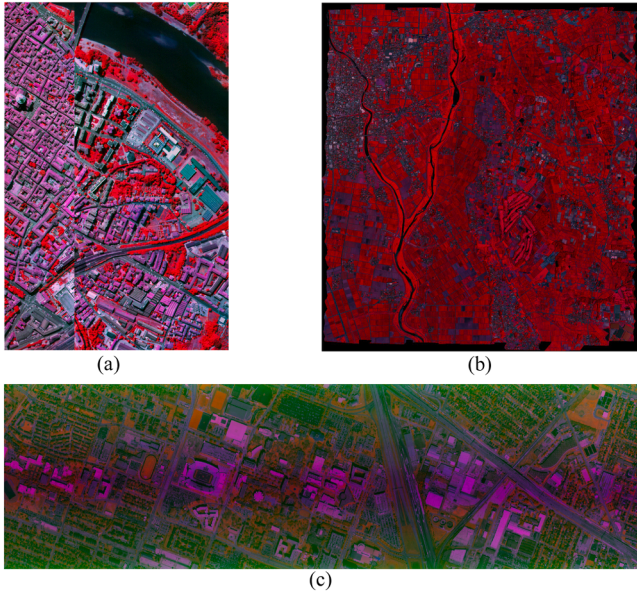


Fig. 5. The false colors images from different datasets. (a) Pavia Centre. False colors are made up of bands 100, 10, and 50. (b) Chikusei. False colors are made up of bands 100, 10, and 50. (c) Houston. False colors are made up of bands 30, 49, and 15.

3.1. Experimental setting

3.1.1. Hyperspectral image dataset

- (1) Dataset Pavia Centre: It was captured by the reflective optical system imaging spectrometer sensor. The spatial size of the centre is 1096×715 with 102 spectral bands. Fig. 5(a) shows its false colors.
- (2) Dataset Chikusei: It captured by the Headwall Hyperspec-VNIR-C imaging sensor shows the agricultural and urban areas of Chikusei, covering the wavelength range of 363–1018 nm. The spatial size of the image is 2517×2335 and contains 128 bands. Fig. 5(b) shows its false colors.
- (3) Dataset Houston: The DFC2018 image in the Houston dataset was captured by a compact airborne spectral imager. The spatial size of the image is 1202×4172 and contains 50 spectral bands. Fig. 5(c) shows its false colors.

3.1.2. Evaluation indicators

In terms of image quality evaluation, three common indicators are used: PSNR, MS-SSIM, and SAM.

- (1) PSNR: It is an objective indicator that evaluates the quality of reconstruction by calculating the difference in pixels in an image, with higher values indicating less distortion. It can be expressed as

$$\text{PSNR}(X, \hat{X}) = \frac{1}{C} \sum_{i=1}^C 10 \log_{10} \left(\frac{\max^2(X^i)}{\text{MSE}_i} \right) \quad (14)$$

Here, MSE stands for the mean square error. $\max^2(X^i)$ stands for the square of the largest pixel in band i . C stands for bands.

- (2) MS-SSIM: It is an indicator that evaluates visual quality by analyzing the brightness, contrast, and structural differences of an image at multiple scales. And the closer its value is to 1, the higher the structural similarity. It can be expressed as

$$D_{\text{MS-SSIM}} = 1 - \prod_{m=1}^M \left(\frac{2\mu_X \mu_{\hat{X}} + C_1}{\mu_X^2 + \mu_{\hat{X}}^2 + C_1} \right)^{\alpha_m} \left(\frac{2\sigma_{X\hat{X}} + C_2}{\sigma_X^2 + \sigma_{\hat{X}}^2 + C_2} \right)^{\zeta_m} \quad (15)$$

Here, M stands for different resolutions. μ_X and $\mu_{\hat{X}}$ stand for the mean of the original hyperspectral data and the reconstructed hyperspectral data. σ_X and $\sigma_{\hat{X}}$ stand for the standard deviation of the original hyperspectral data and the reconstructed hyperspectral data. $\sigma_{X\hat{X}}$ stands for the covariance between the original hyperspectral data and the reconstructed hyperspectral data. α_m and ζ_m stand for the relative importance between the two terms. C_1 and C_2 are constant terms.

To facilitate a clear comparison of the differences in MS-SSIM, they have been transformed into decibel values. It can be represented as

$$\text{MS-SSIM} = -10 \log_{10}(1 - D_{\text{MS-SSIM}}) \quad (16)$$

- (3) SAM: It serves to assess the quality of reconstruction in terms of spectral characteristics. The smaller the value, the smaller the spectral angle difference, i.e., the higher the reconstruction quality. The formula can be represented as

$$\text{SAM}(X, \hat{X}) = \frac{1}{H \times W} \times \sum_{j=1}^{H \times W} \cos^{-1} \left(\frac{x_j \odot \hat{x}_j}{\|x_j\| \|\hat{x}_j\|} \right) \quad (17)$$

Here, $\odot$ stands for the inner product. $\|\cdot\|$ represents the L_2 norm. x_j

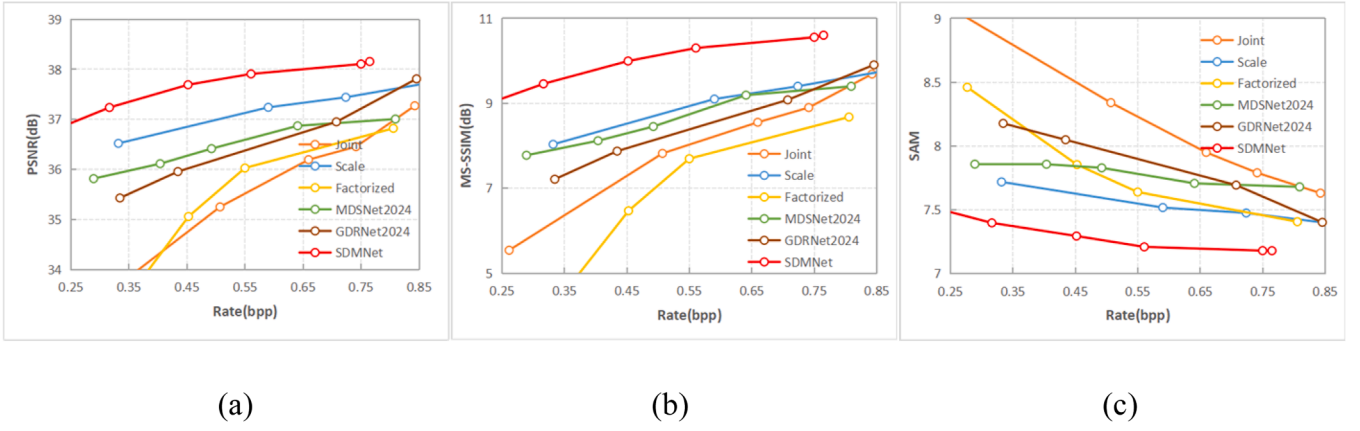


Fig. 6. Rate-distortion performance on Pavia Centre (a) PSNR (b) MS-SSIM (c) SAM.

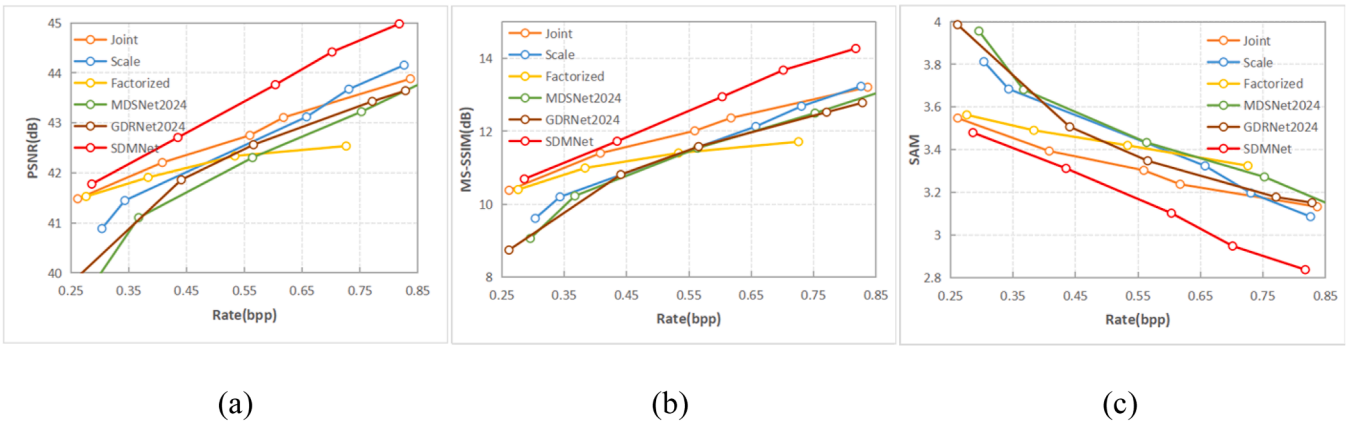


Fig. 7. Rate-distortion performance on Chikusei (a) PSNR (b) MS-SSIM (c) SAM.

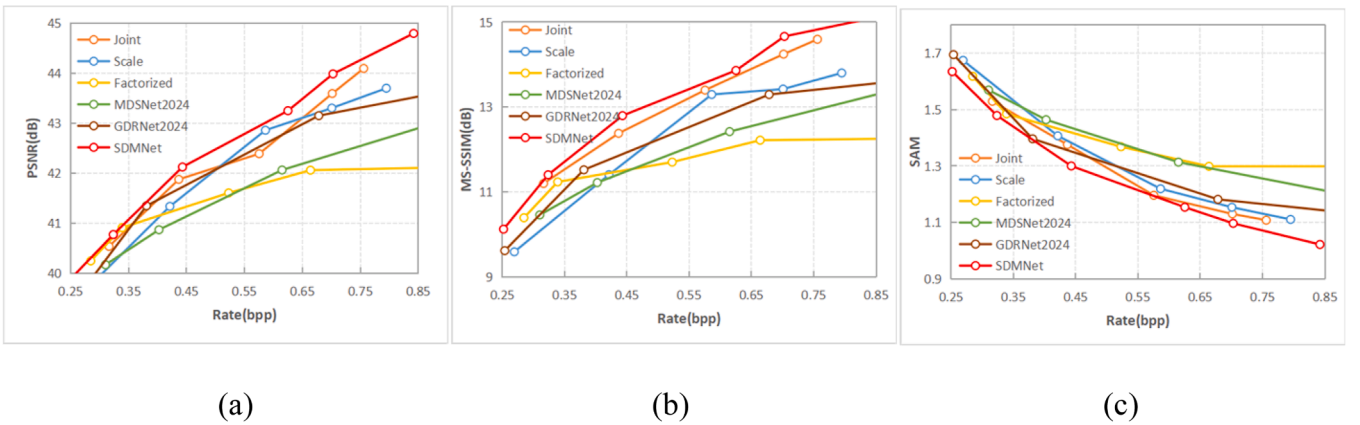


Fig. 8. Rate-distortion performance on Houston (a) PSNR (b) MS-SSIM (c) SAM.

and $\hat{x}_j$ represent the j th pixel of the original hyperspectral image and the reconstructed hyperspectral image, respectively.

3.1.3. Experimental environment and parameter settings

The datasets are randomly cropped into blocks. And the size of blocks is 256×256 . First, 10 % of these blocks are selected as the test set. This ensures that the data used for training and testing do not overlap, avoiding the possibility of data leakage. Second, the remaining parts are divided into training and validation sets in a ratio of 9:1. It should be noted that the cropping stride used here is 64. The minimum number

(50) of spectral bands of the DFC2018 is selected as the smallest unit of spectral grouping to facilitate the unified training of the model. And the partitioning strategy for the three datasets is the same. In addition, no normalization was applied during the creation of the datasets. However, when the model loads the data, Min-Max normalization is used to facilitate faster convergence of the model. SDMNet is developed using PyTorch. In this paper, the commonly used values for the penalty coefficient λ are [0.019, 0.033, 0.065, 0.19, 0.42, 0.65]. Adam is used as an optimizer. This model employs two optimizers: the main optimizer, situated within the main network, and the auxiliary optimizer, posi-

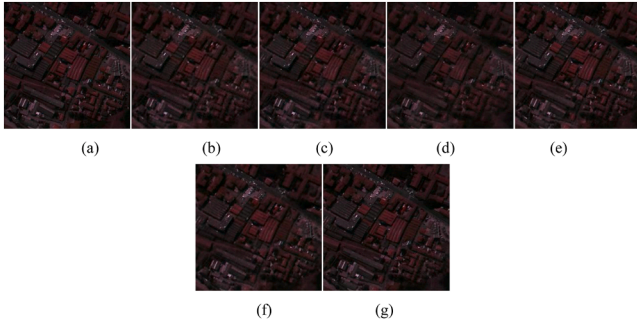


Fig. 9. Visual comparison of the reconstructed hyperspectral images on the Pavia Centre. The false color here is composed of bands 49, 5 and 19. (a) Original image (b) Joint (bpp:0.26; PSNR:33.05; MS-SSIM:5.54; SAM:9.04) (c) Scale (bpp:0.26; PSNR:33.18; MS-SSIM:5.65; SAM: 8.75) (d) Factorized (bpp:0.26 PSNR:32.15; MS-SSIM:3.02; SAM:8.48) (e) MDSNet2024 (bpp:0.25; PSNR:35.18; MS-SSIM:6.95; SAM:7.98) (f) GDRNet2024 (bpp:0.25; PSNR:34.40; MS-SSIM: 5.91; SAM:8.38) (g) SDMNet (bpp:0.25; PSNR:36.92; MS-SSIM:9.10; SAM:7.48).

tioned between the hyper encoder and the hyper decoder. The main optimizer begins with an initial learning rate of 10^{-4} . Throughout the network training, the optimal weights of SDMNet are preserved once the learning rate diminishes to 10^{-5} . The auxiliary optimizer starts with an initial learning rate of 10^{-3} . A batch size of 8 is utilized during training.

3.2. Rate distortion performance

PSNR, MS-SSIM and SAM were employed to evaluate these models. Figs. 6,7, and 8 show the PSNR, MS-SSIM, and SAM rate distortion performance curves for all models on the Pavia Centre, Chikusei, and Houston datasets, respectively. Factorized exhibits poor performance on all datasets. This is mainly due to the weak feature capture ability caused by its overly simplistic basic framework. Joint showed good performance on the Houston dataset, but poor performance on both Pavia Centre and Chikusei. This illustrates its poor robustness. Scale, MDSNet2024, and GDRNet2024 show good rate distortion performance, mainly due to their efficient basic framework and unique attention mechanism, but their performance is still much lower than SDMNet. The proposed SDMNet achieves the best performance on all datasets, which not only shows the superior compression performance, but also demonstrates the strong robustness. As shown in Table 1, under high compression ratio conditions(0.25bpp), the proposed SDMNet has an 11.7 % performance advantage in PSNR compared to the Joint method

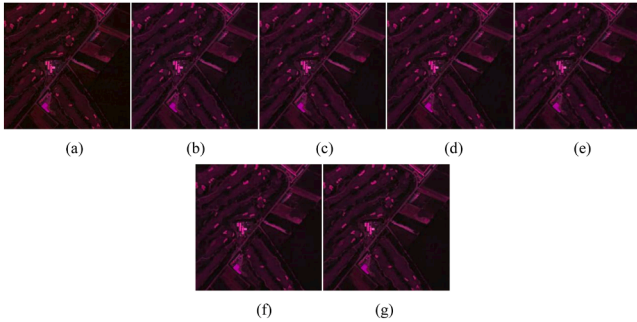


Fig. 10. Visual comparison of the reconstructed hyperspectral images on the Chikusei. The false color here is composed of bands 49, 2, and 30. (a) Original image (b) Joint (bpp:0.26; PSNR: 41.66; MS-SSIM:10.35; SAM:3.49) (c) Scale (bpp:0.26; PSNR:40.14; MS-SSIM:8.77; SAM:3.96) (d) Factorized (bpp:0.25; PSNR:41.07; MS-SSIM:9.81; SAM:3.65) (e) MDSNet2024 (bpp:0.25; PSNR:39.39; MS-SSIM:8.31; SAM:4.03) (f) GDRNet2024 (bpp:0.26; PSNR:39.92; MS-SSIM:8.73; SAM:3.98) (g) SDMNet (bpp:0.25; PSNR:41.35; MS-SSIM:10.15; SAM:3.51).

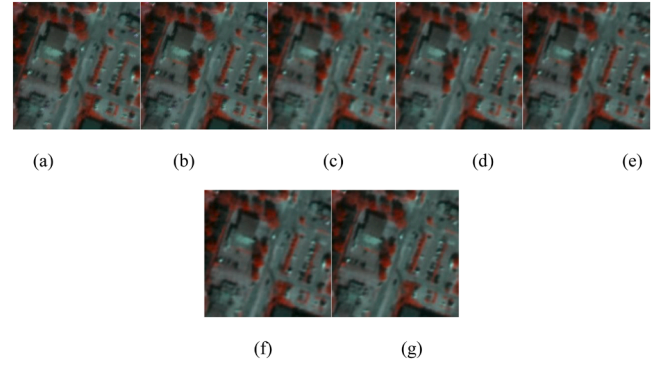


Fig. 11. Visual comparison of the reconstructed hyperspectral images on the Houston. The false color here is composed of bands 30, 15, and 20. (a) Original image (b) Joint (bpp:0.20; PSNR: 37.26; MS-SSIM:9.18; SAM:1.94) (c) Scale (bpp:0.20; PSNR:36.28; MS-SSIM:6.66; SAM:2.36) (d) Factorized (bpp:0.22 PSNR:37.14; MS-SSIM:7.44; SAM:2.11) (e) MDSNet2024 (bpp:0.21; PSNR:38.15; MS-SSIM:8.61; SAM:2.35) (f) GDRNet2024 (bpp:0.21; PSNR:36.68; MS-SSIM:7.68; SAM:2.35) (g) SDMNet (bpp:0.22; PSNR:39.44; MS-SSIM:9.93; SAM:1.68).

on Pavia Centre. In addition, Tables 2 and 3 also show the specific results on two other datasets. This superior performance also greatly proves the effectiveness of preserving spectral differences and refining feature extraction to improve hyperspectral compression performance. In addition, this performance is further validated through ablation studies (Section 3.4), confirming the synergistic effects of SDPH and MPFM, while complexity analysis (Section 3.5) shows low parameters.

3.3. Visualization comparison experiment of reconstructed images

To verify the visual effect of reconstructed images, some visualization comparison experiments were conducted. Fig. 9 shows the reconstructed hyperspectral images of all methods on the Pavia Centre at 0.25bpp. Fig. 10 shows results of all methods on the Chikusei at 0.25bpp. Fig. 11 shows results of all methods on the Houston at 0.20bpp. Fig. 9 shows that Joint, Scale and Factorized all have common artifacts and noise problems, which cause blurring effects in local areas of the image. In addition, it can be observed that results of these three comparison methods appear rough in the transition processing between pixels, which leads to distortion. MDSNet2024 and GDRNet2024 can retain some texture features, but their localized regions appear overly smoothed, resulting in the loss of intricate details. By comparison, it is found that SDMNet achieves the best visualization effect. In addition, SDMNet also achieved good visualization effects on Chikusei and Houston, which also verifies the robustness of SDMNet. This fully verifies that SDPH and MPFM can efficiently extract important features from hyperspectral data.

3.4. Ablation experiments

Some ablation experiments were conducted to verify the effectiveness of SDPH and MPFM. The ablation experiment results on the Pavia Centre, the Chikusei, and the Houston are shown in Figs. 12-14. Baseline represents the basic model, which is a simple VAE model. SDMNet (SDPH) represents the model with SDPH added to the baseline. SDMNet (MPFM) represents the model with MPFM added to the baseline. SDMNet represents the entire network, including the baseline, SDPH, and MPFM. It can be seen from Figs. 12,13, and 14 that after adding SDPH and MPFM, the network performs better than the Baseline. This fully demonstrates that it is of great significance to retain spectral differences and to refine the extraction of complex hyperspectral features. SDMNet has achieved the best performance, which fully demonstrates that the modules are compatible and can work together efficiently.

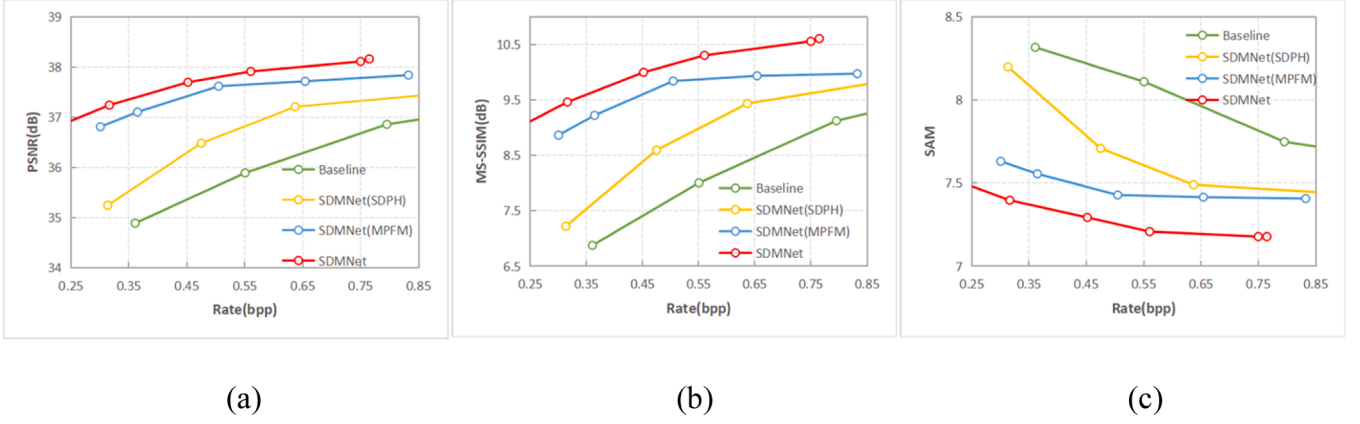


Fig. 12. Ablation experiment results on the Pavia Centre. (a) PSNR, (b) MS-SSIM, (c) SAM.

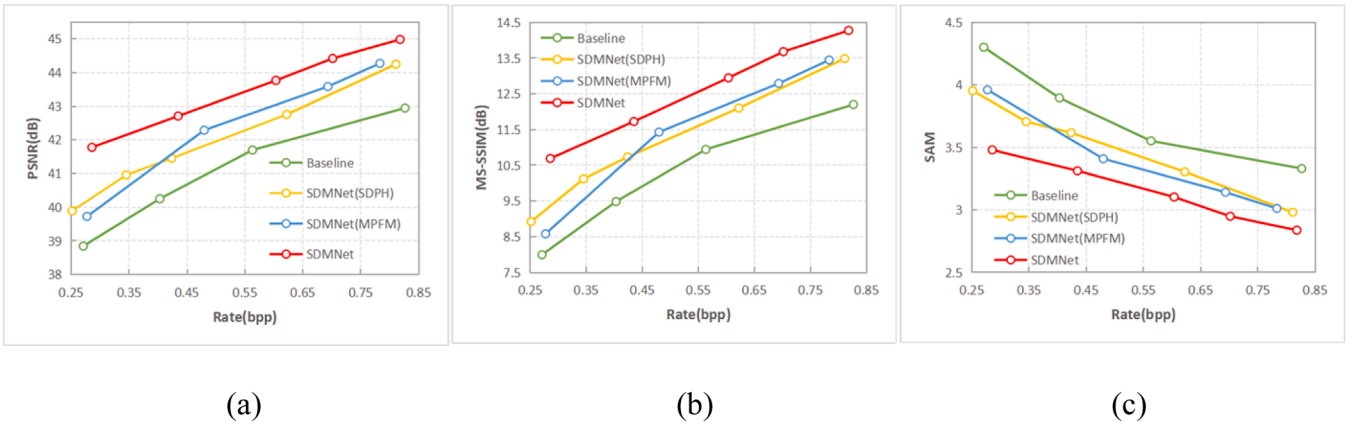


Fig. 13. Ablation experiment results on the Chikusei. (a) PSNR, (b) MS-SSIM, (c) SAM.

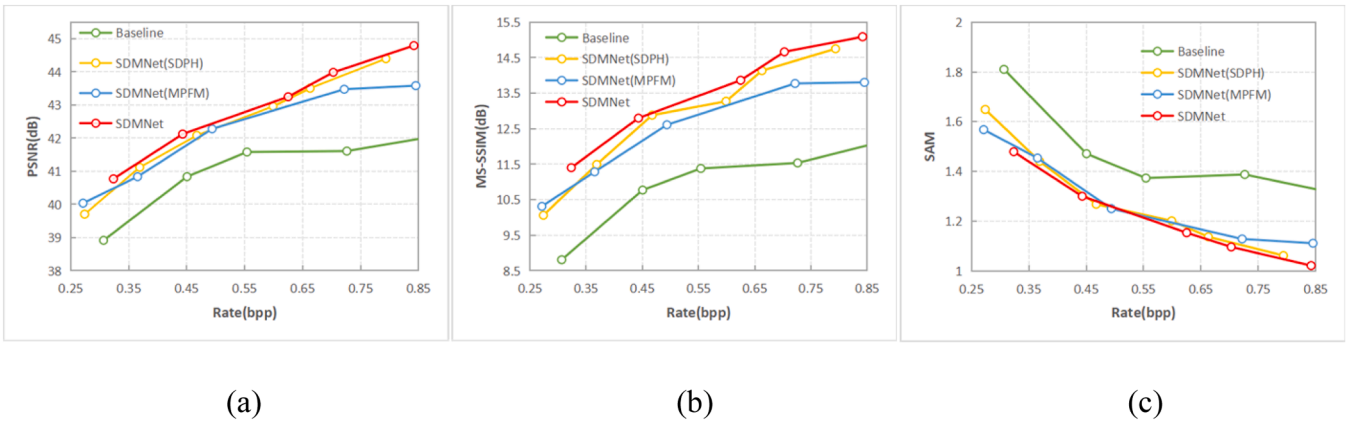


Fig. 14. Ablation experiment results on the Houston. (a) PSNR, (b) MS-SSIM, (c) SAM.

Table 4

Complexity parameters for different methods.

	Joint	Scale	Factorized	MDSNet2024	GDRNet2024	SDMNet
Parameter	12.50M	10.36M	6.01M	8.82M	11.32M	6.30M
FLOPs	45.52G	44.97G	44.43G	18.09G	16.76G	21.54G
GPU Memory	3.5GB	3.1GB	2.3GB	4.4GB	5.7GB	5.3GB
Compression time	0.7901s	0.0179s	0.0069s	0.0244s	0.0155s	0.0081s
Reconstruction time	1.5102s	0.0158s	0.0074s	0.0206s	0.0115s	0.0077s

3.5. Complexity analysis

This paper conducts a complexity analysis. The input image format is $50 \times 256 \times 256$. As shown in Table 4, SDMNet has a great advantage in the number of parameters. By comparing Parameter, it can be found that SDMNet has achieved the second lowest number of parameters, which is only 50.4 %, 60.8 %, 104.8 %, 71.4 % and 55.7 % of Joint, Scale, Factorized, MDSNet2024 and GDRNet2024 respectively. By comparing FLOPs, it can be found that SDMNet still has the third lowest number of parameters. Compared with GPU Memory, SDMNet consumes more resources. This is because some parallel modules are used in the network. Last, it can be found that the time consumed by SDMNet is slightly slower than the fastest method. However, the PSNR, MS-SSIM and SAM of SDMNet are much better than other methods. This fully verifies the efficiency of SDMNet.

4. Conclusions

In this paper, a novel SDMNet was proposed for the compression of hyperspectral images, which preserves the differences between hyperspectral spectra while achieving refined extraction of various features. Specifically, we use SDPH to efficiently align the frequency-domain features in the original hyperspectral image and the reconstructed hyperspectral image. Then, we further use MPFM to perform refined extraction of multiple features. These specific designs enable the proposed model to demonstrate better compression performance both in terms of specific metrics and visualizations. However, the proposed SDMNet has a major limitation: the GPU memory usage during model operation reaches as high as 5.3GB, which poses a severe challenge for deploying the model on some lightweight platforms. The main reason for this issue is that the model uses many multi-branch structures, resulting in a large amount of data that must be processed in parallel. In future work, we will further optimize the model, reduce the number of parameters, and strive to achieve efficient real-time compression of hyperspectral images. In addition, our exploration of Mamba in the field of hyperspectral compression is preliminary, and we will further explore Mamba and its variants in the field of compression.

Appendix

Tables A1 and A2 show the specific parameters of C blocks 1-3, R blocks 1-3, the Hyper encoder, and the Hyper decoder.

Table A1
C block 1-3 and R block 1-3.

Block	First layer	Second layer
C block 1	Conv2D 3×3 N/4 N/2 2↓	GDN
C block 2	Conv2D 3×3 N/2 3N/4 2↓	GDN
C block 3	Conv2D 3×3 3N/4 N 2↓	GDN
R block 1	TransposedConv2D 3×3 N/2 N/4 2↑	IGDN
R block 2	TransposedConv2D 3×3 3N/4 N/2 2↑	IGDN
R block 3	TransposedConv2D 3×3 N 3N/4 2↑	IGDN

Table A2
Hyper encoder and Hyper decoder.

	Hyper encoder	Hyper decoder
Layer1	Conv2D 3×3 N N 1	TransposedConv2D 3×3 N N 2↑
Layer2	RELU	RELU
Layer3	Conv2D 3×3 N N 2↓	TransposedConv2D 3×3 N N 2↑
Layer4	RELU	RELU
Layer5	Conv2D 3×3 N N 2↓	Conv2D 3×3 N N 1
Layer6	-	RELU

CRediT authorship contribution statement

Kaijie Shi: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cuiping Shi:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Weiwei Sun:** Visualization, Validation, Project administration. **Liguo Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Cuiping Shi reports financial support was provided by National Natural Science Foundation of China under Grant 42271409, and provided by Science and Technology Plan Project of Huzhou under Grant 2024GZ36. Weiwei Sun reports financial support was provided by Project National Key R&D Program of China under Grant 2024YFF1400900, and provided by Zhejiang Province "Pioneering Soldier" and "Leading Goose" R&D Project under Grant 2023C01027, and and provided by the Key Technology Breakthrough Plan Project of Science and Innovation Yongjiang 2035 under Grant 2024Z262 and 2022Z032, and provided by Youth Scientist Project National Key R&D Program of China under Grant 2023YFF1305600. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported in part by the National Natural Science Foundation of China (42271409), in part by the Science and Technology Plan Project of Huzhou under Grant 2024GZ36, and in part by Project National Key R&D Program of China under Grant 2024YFF1400900, and in part by Zhejiang Province "Pioneering Soldier" and "Leading Goose" R&D Project under Grant 2023C01027, and in part by the Key Technology Breakthrough Plan Project of Science and Innovation Yongjiang 2035 under Grant 2024Z262 and 2022Z032, and in part by Youth Scientist Project National Key R&D Program of China under Grant 2023YFF1305600.

In Tables A1 and A2, N stands for channels and is usually set to 256, $2\downarrow$ stands for downsampling stride of 2, $2\uparrow$ stands for upsampling stride of 2, and RELU stands for the linear rectification function. GDN represents generalized split normalization function and IGDN represents its inverse operation.

Table A3
Complexity Analysis of Mamba and Transformer.

	Parameter	FLOPs	Times
MPFM(Mamba)	2.20 G	0.82 M	1.56ms
MPFM (Transformer)	2.37 G	0.83 M	2.44ms

In Table A3, we take MPFM out separately for complexity testing. MPFM (Mamba) is the original module in the paper, while MPFM (Transformer) replaces the Mamba in the original module with a Transformer. The 'Times' here refers to the time consumed by the network to perform a hundred forward passes on the module. The three indicators show that, in this module, Mamba outperforms Transformer both in terms of parameters and running efficiency.

Data availability

Data will be made available on request.

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